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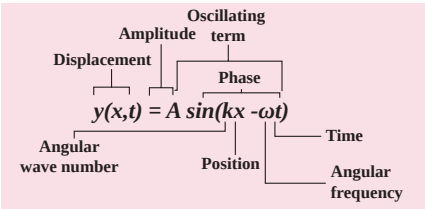
$a = \frac{3\pi}{2}$ and $a = \pi$, the function $y = \sin(x-a)$ shifts towards right. Further, we can take $a = vt$ and $v = \frac{\pi}{4}$, and sketching for different times $t = 0s$, $t = 1s$, $t = 2s$ etc., we once again observe that $y = \sin(x-vt)$ moves towards the right. Hence, $y = \sin(x-vt)$ is a travelling (or progressive) wave moving towards the right. If $y = \sin(x+vt)$ then the travelling (or progressive) wave moves towards the left. Thus, any arbitrary function of type $y = f(x-vt)$ characterising the wave must move towards right and similarly, any arbitrary function of type $y = f(x+vt)$ characterizing the wave must move towards left.

EXAMPLE 11.13

Check the dimensional of the wave $y = \sin(x-vt)$. If it is dimensionally wrong, write the above equation in the correct form.

Solution

Dimensionally it is not correct. we know that $y = \sin(x-vt)$ must be a dimensionless quantity but $x-vt$ has dimension. The correct equation is $y = \sin(k\,x-\omega t)$, where k and ω have the dimensions of inverse of length and inverse of time respectively. The sine functions and cosine functions are periodic functions with period 2π. Therefore, the correct expression is $y = \sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t\right)$ where λ and T are wavelength and time period, respectively. In general, $y(x,t)=\mathrm{A}\sin(k\,x-\omega t)$.



11.6.3 Graphical representation of the wave

Let us graphically represent the two forms of the wave variation

(a) Space (or Spatial) variation graph

(b) Time (or Temporal) variation graph

(a) Space variation graph

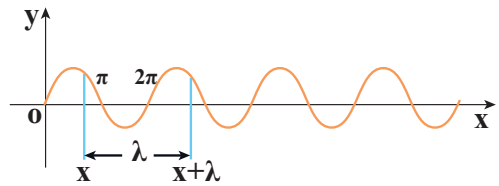


Figure 11.24 Graph of sinusoidal function $y = A\sin(kx)$

By keeping the time fixed, the change in displacement with respect to x is plotted. Let us consider a sinusoidal graph, $y = A\sin(kx)$ as shown in the Figure 11.24, where k is a constant. Since the wavelength λ denotes the distance between any two points in the same state of motion, the displacement y is the same at both the ends

$y = x$ and $y = x + \lambda$, i.e.,

$$y = \mathrm{A}\sin(kx) = \mathrm{A}\sin(k(x+\lambda)) = \mathrm{A}\sin(kx + k\lambda) \qquad (11.38)$$

The sine function is a periodic function with period 2π. Hence,

$$y = \mathrm{A}\sin(kx + 2\pi) = \mathrm{A}\sin(kx) \qquad (11.39)$$

Comparing equation (11.38) and equation (11.39), we get

$$kx + k\lambda = kx + 2\pi$$

This implies

$$k = \frac{2\pi}{\lambda}\ \mathrm{rad\ m^{-1}} \qquad (11.40)$$

where k is called wave number. This measures how many wavelengths are present in 2π radians.